Rohini Das

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EDUCATION

Carnegie Mellon University | *Master of Computational Data Science (GPA 3.99/4.00)* **Pittsburgh, USA | December 2026**
Relevant Courses: ML in Production, Cloud Computing, Advanced NLP, Multimodal ML, Interactive Data Science (TA)

VIT University | *Bachelors of Technology, Computer Science Engineering (GPA 3.985/4.00)* **Vellore, IN | August 2024**
Relevant Courses: Splz. in Bioinformatics, Data Structures and Algorithms, Object-Oriented Programming, Deep Learning

SKILLS

Programming: Python, SQL, R, Java, Scala, Django, Flask, Node.js, Semantic Kernel C# .NET SDK, Neo4j
Models: Transformers, Diffusion Models, Vision-Language Models, BERT, ResNet,CNNs, RNNs, Knowledge Graphs
Machine Learning Engineering: PyTorch, TensorFlow, Keras, Scikit-learn, Feature Engineering, XGBoost, Hugging Face
Generative AI: LLMs, MCP, AI Agents, LangGraph, LangChain, RAG, Prompt Engineering, AzureOpenAI, Ollama, FAISS
Cloud & Deployment: AWS (EKS, Lambda, S3, EC2), GCP, Azure, Minikube, Docker, Kubernetes, Git CI/CD, FastAPI
Data Science Analytics: Pandas, NumPy, A/B Testing, Matplotlib, Seaborn, Streamlit, PySpark, Spark, Hadoop, Kafka, MongoDB

EXPERIENCE

Qualcomm *Santa Clara, CA*
AI Engineering Intern *May 2026 - Aug 2026*

- Building Agentic AI tools for WLAN systems engineering: **telemetry evaluation**, structured information retrieval from extended research standard documentation and end-to-end **automation workflows** with full traceability.

Siemens Foundational Technologies *Bangalore, IN*
Artificial Intelligence Engineer *Aug 2024 - Aug 2025*

- Designed and deployed the publication pipeline and **MCP Registry** within **Agent-2-Agent (A2A) framework** for *Siemens **Industrial Copilot** for Operations* © achieving low-latency real-time validation at ~3s turnaround.
- Engineered a production-grade, modular **Natural Language to SQL (NL2SQL)** agentic system with persistent memory, dynamic state management, and large-scale SQLServer querying via **LangGraph;** reduced internal MIS data retrieval from 2–3 hours to on-demand, eliminating a critical bottleneck.

Business Intelligence Intern *Mar 2024 - July 2024*

- Proposed an **AI-based log analysis** of Non Functional Requirements testing of *Siemens Simatic HMI Panels*, filed as an ***Invention Disclosure*** No. 2024E14797 IN. Reduced development time **by 10+ hours** per fortnight; increased bug redressal quality for Screen Freeze regressions.
- Built Digital GenAI-Assistant, a **full-stack web app** automating Quality Management Review analysis by extracting **multi-source tabulated data** from MS Excel, PowerPoint, Azure DevOps. Deployed **GenAI-mediated negotiation** agents for live multi-user dispute resolution, replacing 1-hour+ quarterly meetings with a fair, structured **LLM-as-a-judge** evaluation system.

Computer Vision Research Intern *Sep 2022 - Mar 2023*

- Incorporated **YOLOv5 with Ultralytics weights** in a self-supervised neural network for segmenting printed text from bank checks with mixed handwritten and printed content. Published in **IEEE ICACCS 2023**, the paper won the **Best Paper Award** for reducing an 11-step bank check collection process to 5 simple steps with very little human intervention.

PROJECTS & RESEARCH

NL2SQL Agent Benchmarking on Spider 2.0 *Capstone Research | CMU | Jan 2026 - Dec 2026*

- Designing a **multi-agent** NL2SQL system evaluated against the **Spider 2.0 benchmark**, served via **vLLM** on SLURM.
- It integrates enriched semantic schema retrieval, **explicit query planning**, sub-agent execution, SQL validation and self-correction loops; evaluated across GLM-4.7-Flash-AWQ, Gemini 3 Pro, and GPT-5.5.

Real-Time Movie Recommendation Platform *Course Project | CMU | Jan 2026 – Present*

- Trained PyTorch **matrix-factorization** model on 236K users × 20K movies from live **Kafka** streams; shipped zero downtime Dockerized dual-instance ONNX serving with automated retraining via GitHub Actions, a hybrid cold-start fallback at ~7.6–10.4 req/s.
- E2E stack: Offline (RMSE/MAE), online (**Watch Precision@20**) evaluation, Prometheus-Grafana dashboard tracking latency, drift. Validated a **cost-aware reranker** that cut recommendation license cost by 93% ($6.01 → $0.42) via a full A/B framework with Welch's t-test and Mann-Whitney U-test.

FedHealth: Collaborative ML for ICU Mortality Prediction *Capstone Project | VIT | Jan 2024 - July 2024*

- Created a privacy-preserving, decentralized mortality prediction model using **PyTorch**, **Flower.ai** with ICU-EHR data from the MIMIC-IV database to identify high mortality patients based on medical history of Prostate Cancer, comorbidities, medication.
- Applied **FedAvg, FedProx** federated ML algorithms, achieving an **F1 Score ~0.94** and **95.3% accuracy** outperforming a sequential ANN baseline (F1: 0.82) by a significant margin.

Automated Face Detection *Team Project | VIT | Aug 2023 - Dec 2023*

- Led a 4-person team to build a **face detection**, **recognition** pipeline in **TensorFlow** via **transfer learning ResNet50** on the AT&T Faces, FaceScrub DBs by **hyperparameter tuning** during **segmentation**; fine-tuned model at **98.7% val accuracy v/s baseline 95%**.